The Game Theory of Bluffing in B.S.

Leo Cambisaca, Ben Pazderak, Sahana Savarkar

09-05-25

Abstract

This paper analyses strategic behavior in the bluffing card game "B.S." (also known as "Cheat" or "Bluff") in which players try to discard all of their cards while concealing the truth about the cards they play. We are investigating the best responses to mixed strategies and identifying the dominant strategy for the players by simulating repeated games. Payoff matrices represent the expected outcomes of "calling" versus "not calling" a suspected bluff.

Introduction

Our initial payoff matrix is a 2×2 matrix as follows, where:

	C	**D**
T	(2, -2)	(1, 0)
B	(-3, 2)	(3, -1)

C: call a suspected bluff,
D: don't call a suspected bluff,
T: telling the truth,
B: bluffing

Best Responses against Mixed Strategies

In order to evaluate optimal behaviour in B.S., we analyse best responses for when players randomise their choices without committing to any single pure strategy.
We isolate the row and column players' payoffs and denote these matrices as **A** and **B** respectively, and the mixed strategies for row and column players are denoted by σ_r and σ_c respectively.

$$\mathbf{A}\sigma_r^T = \begin{pmatrix} 2 & 1 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} x \\ 1-x \end{pmatrix} = \begin{pmatrix} x+1 \\ -6x+3 \end{pmatrix}$$

$$\sigma_c \mathbf{B} = \begin{pmatrix} y & 1-y \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} -4y+2 & y-1 \end{pmatrix}$$

Setting the row and column payoffs equal, we find that the Nash equilibria are $\frac{2}{7}$ and $\frac{3}{5}$ respectively, i.e.

- if $x < \frac{2}{7}$, then the row player prefers T (truth),
  if $x > \frac{2}{7}$, then the row player prefers B (bluff),
  and if $x = \frac{2}{7}$, they are indifferent and can randomise due to equal average payoffs.
- if $y < \frac{3}{5}$, the column player prefers C (call),
  if $y > \frac{3}{5}$, the column player prefers D (don't call),
  and if $y = \frac{3}{5}$, they are indifferent and can randomise due to equal average payoffs.

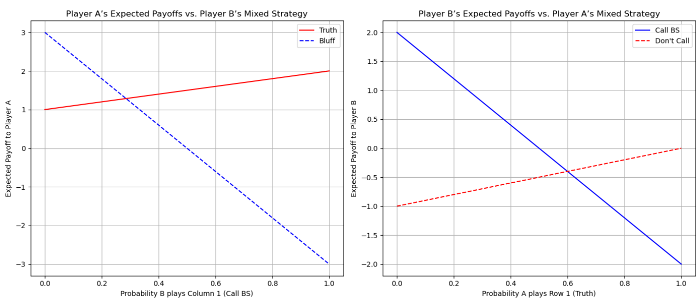

Figure 1: Mixed Strategies and their Expected Payoffs

Results

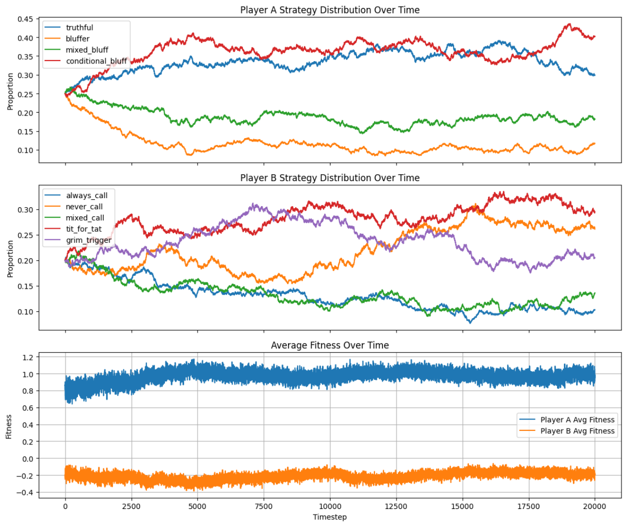

Figure 2: Co-evolving Moran Process with Mutation = 0.1 for Bluffing and Calling Strategies

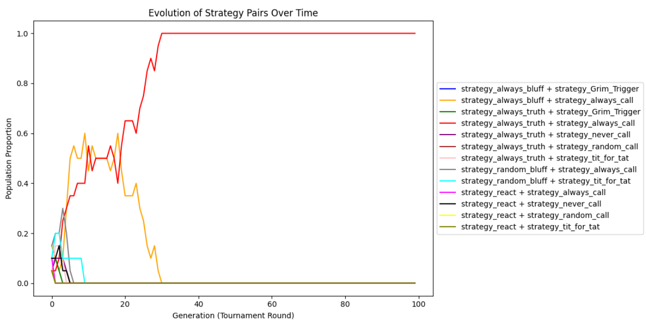

Figure 3: Proportion of Strategies being used by Row & Column players Over Time

Interpretation

- **Bluffing strategy:** We observe that conditional bluffing (bluffing after a no-call last turn) and entirely truthful strategies have higher proportions compared to 50/50 mixed bluffing and only bluffing. Though this is a single simulation and can change due to randomised mutation, this seems consistent across several iterations, and demonstrates that strategies with higher proportions of truth-telling tend to be used more in the long run.
- **Calling strategy:** The strategies with the highest proportion over time are tit-for-tat, never calling, and the grim trigger (calling consistently after a player has bluffed previously). These make sense in the context of the strategies evolved for the bluffing player, which tend to tell the truth more of the time.
- **Average fitness:** Notably, the average fitness for the calling player stays below 0, indicating that they don't often call out bluffs, and would rather not call to avoid a larger negative payoff if the bluffer was telling the truth.

Interpretation

- This model closely resembles repeated-games replicator dynamics (where the evolution of strategies is determined by the relative payoff matrix). In this implementation, we use score-based selection to grow strategies that perform better.
- These evolving graphs highlight how the population strategy pairs converge into a pair of "best strategy pairs" for a distinct game, as having different initial strategies or hands leads to different stable outcomes. This reflects the true idea of B.S., where a player's success is determined by the other players' strategies.

Conclusions

We notice that in an ideal world, strategies involving mostly truth-telling tend to be better for bluffing players, while non-calling strategies or reactive strategies tend to be better for calling players. This tracks with the Nash equilibrium calculated in the Best Responses section. In practice, game strategies evolve depending on how others play, and therefore, we still see a larger amount of bluffing in actual games. Some limitations of this include a single payoff matrix being used, where different weights could influence different play styles. Furthermore, the Moran Process assumes a case where players can always tell the truth, which is not often the case in actual games where players are forced to bluff.

References

[1] Vincent Knight, Owen Campbell, and Marc Harper. *Nashpy: A Python library for the computation of Nash equilibria*. Journal of Open Source Software, 6(63):3065, 2021. Available at: https://doi.org/10.21105/joss.03065

[2] Vincent Knight. *Playing games: A case study in active learning applied to game theory*. MSOR Connections, 14(1):28–38, 2015.